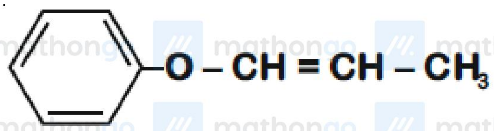
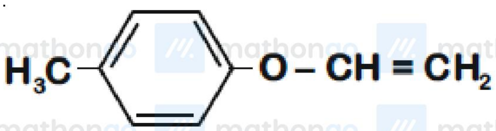
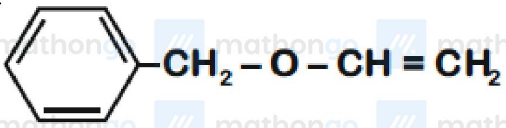
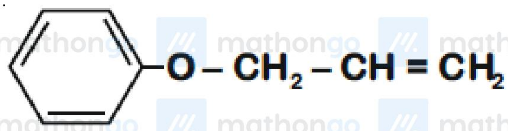
Q1 JEE Main 2020 - 2 September (Evening)

An organic compound 'A' $(C_9H_{10}O)$ when treated with conc. HI undergoes cleavage to yield compounds 'B' and 'C'. 'B' gives yellow precipitate with $AgNO_3$ where as 'C tautomerizes to 'D'. 'D' gives positive iodoform test. 'A' could be

(A) 

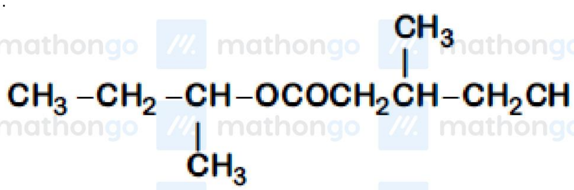
(B) 

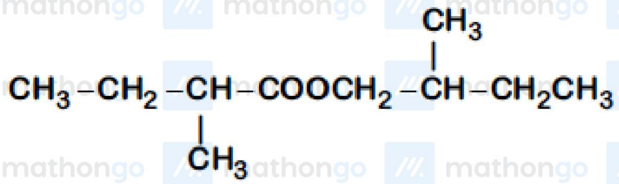
(C) 

(D) 

Q2 JEE Main 2020 - 3 September (Morning)

An organic compound [A], molecular formula $C_{10}H_{20}O_2$ was hydrolyzed with dilute sulphuric acid to give a carboxylic acid [B] and an alcohol [C]. Oxidation of [C] with $CrO_3 - H_2SO_4$ produced [B]. Which of the following structures are not possible for [A]?

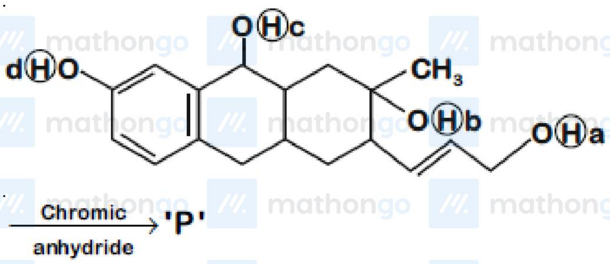
(A) 



(B)
(C) $(CH_3)_3C - COOCH_2C(CH_3)_3$
(D) $CH_3CH_2CH_2COOCH_2CH_2CH_2CH_3$

Q3 JEE Main 2020 - 3 September (Evening)

Consider the following reaction :

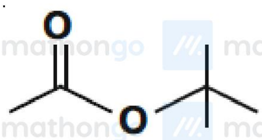


The product 'P' gives positive ceric ammonium nitrate test. This is because of the presence of which of these -OH group(s)?

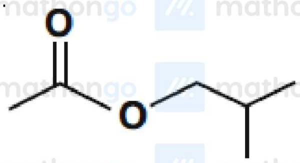
(A) (d) only
(B) (c) and (d)
(C) (b) only
(D) (b) and (d)

Q4 JEE Main 2020 - 4 September (Morning)

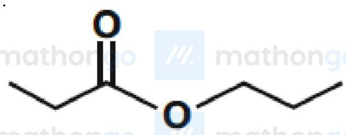
An organic compound (A) (molecular formula $C_6H_{12}O_2$) was hydrolysed with dil. H_2SO_4 to give a carboxylic acid (B) and an alcohol (C). 'C gives white turbidity immediately when treated with anhydrous $ZnCl_2$ and conc. HCl. The organic compound (A) is



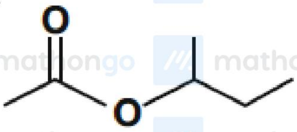
(A)



(B)



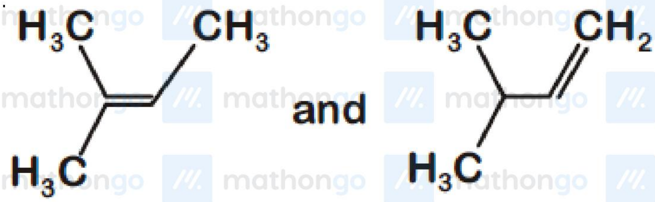
(C)



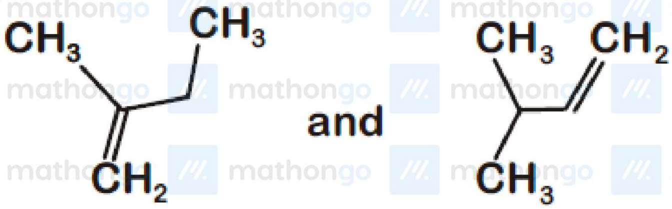
(D)

Q5 JEE Main 2020 - 4 September (Morning)

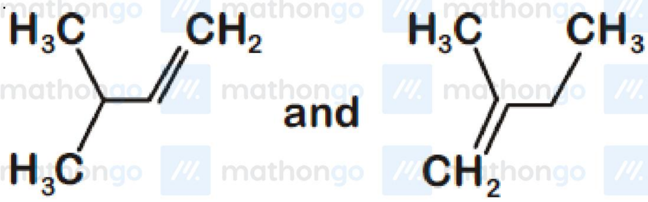
When neopentyl alcohol is heated with an acid, it slowly converted into an 85: 15 mixture of alkenes A and B, respectively. What are these alkenes?



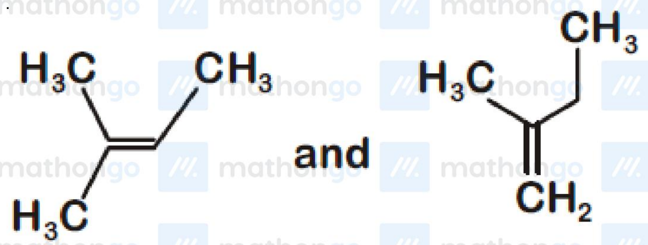
(A)



(B)

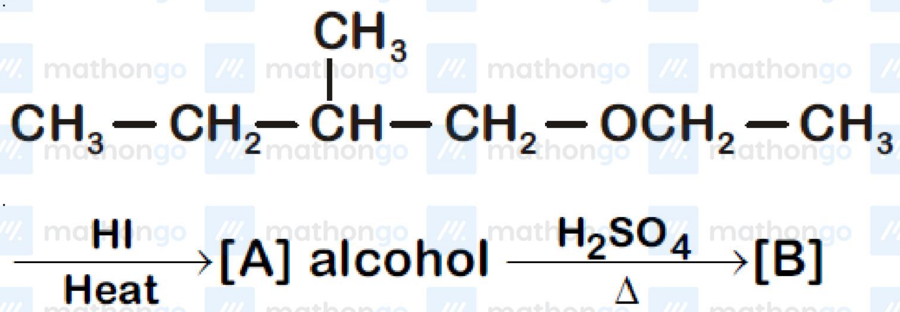


(C)

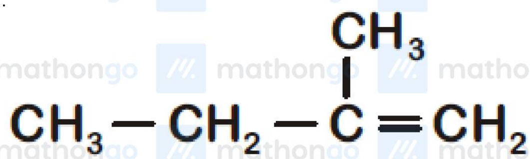
(D) 

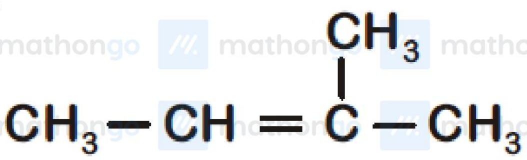
Q6 JEE Main 2020 - 4 September (Evening)

The major product [B] in the following reactions is



(A) $CH_3 - CH_2 - CH = CH - CH_3$

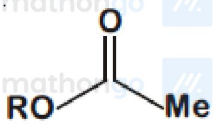
(B) 

(C) 

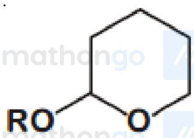
(D) $CH_2 = CH_2$

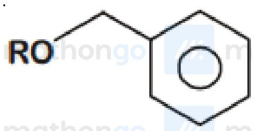
Q7 JEE Main 2020 - 5 September (Morning)

Which of the following derivatives of alcohols is unstable in an aqueous base?


(A)

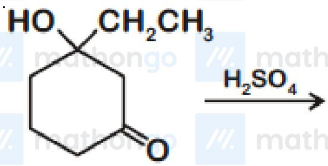
(B) $\mathrm{RO}-\mathrm{CMe}_3$

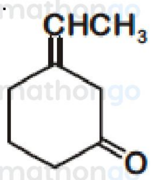

(C)

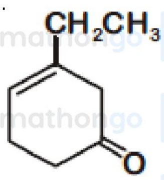

(D)

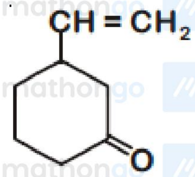
Q8 JEE Main 2020 - 5 September (Evening)

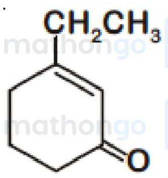
The major product of the following reaction is




(A)


(B)

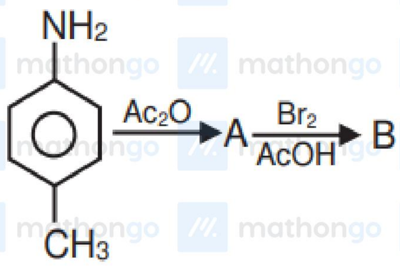
(C) 

(D) 

Q9 JEE Main 2020 - 6 September (Evening)

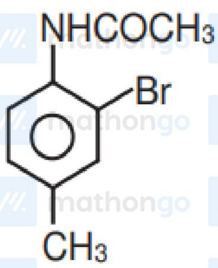
A solution of phenol in chloroform when treated with aqueous NaOH gives compound P as a major product. The mass percentage of carbon in P is ____. (to the nearest integer)
(Atomic mass : $\mathrm{C}=12; \mathrm{H}=1; \mathrm{O}=16$)

Q10 JEE Main 2020 - 7 January (Evening)

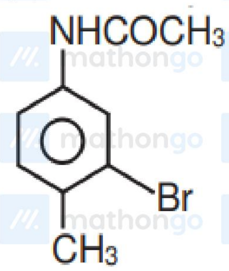


A and B are in the given reaction ?

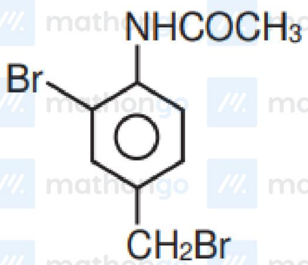
(A)



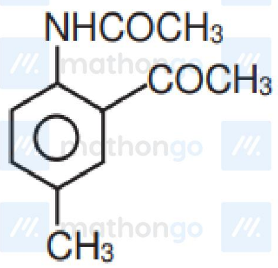
(B)



(C)

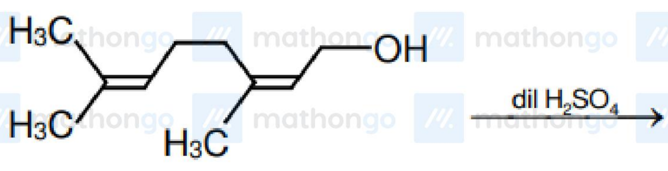


(D)

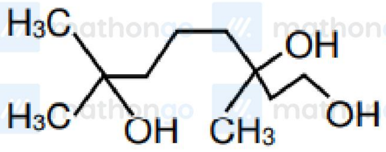


Q11 JEE Main 2020 - 8 January (Morning)

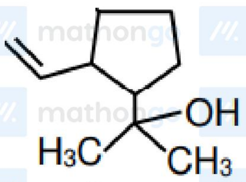
The major product of the following reaction is :



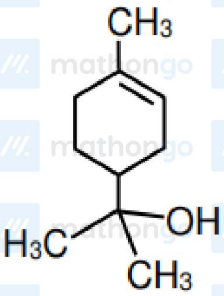
(A)



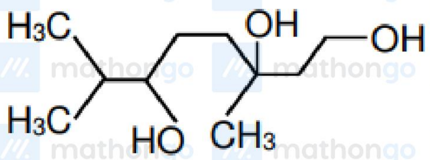
(B)



(C)

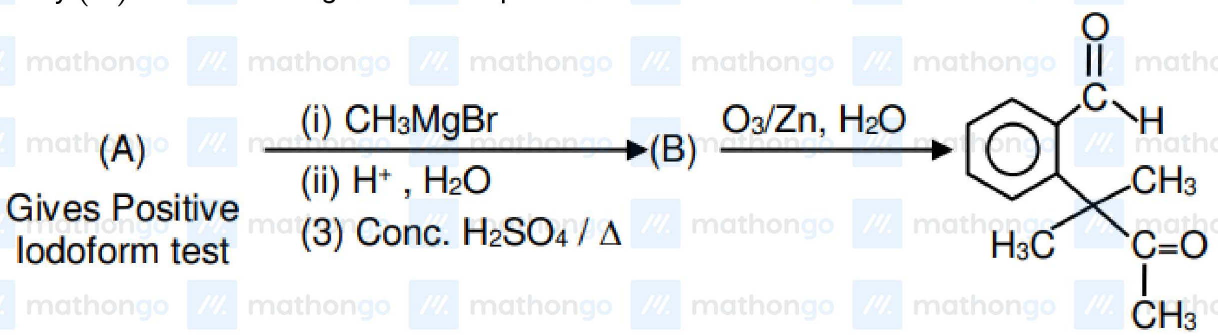


(D)

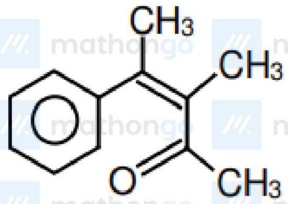


Q12 JEE Main 2020 - 9 January (Morning)

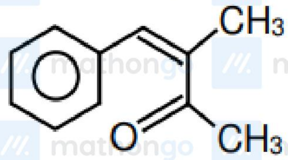
Identify (A) in the following reaction sequence :



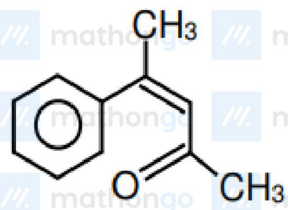
(A)



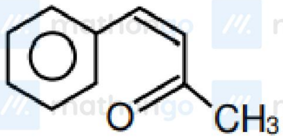
(B)



(C)



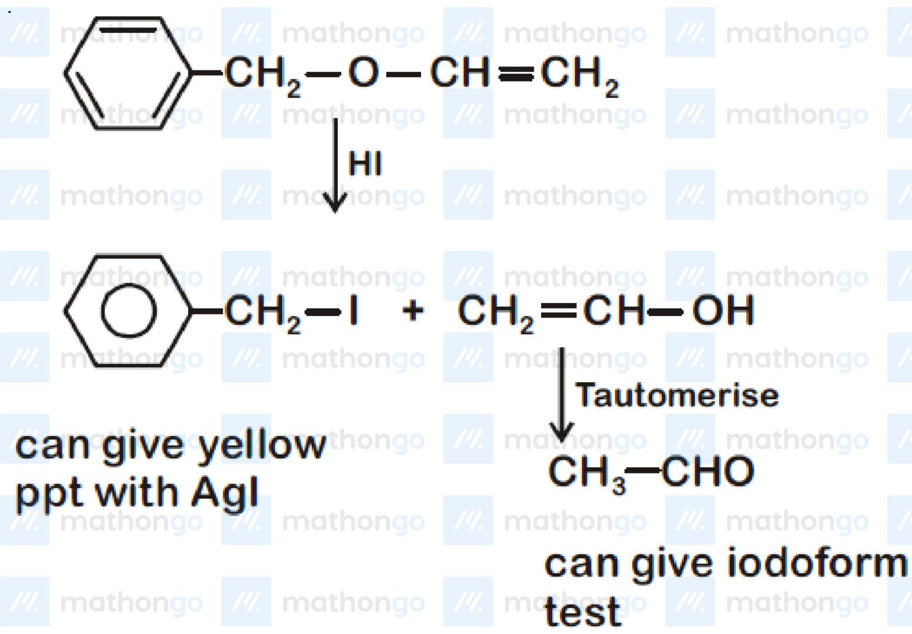
(D)



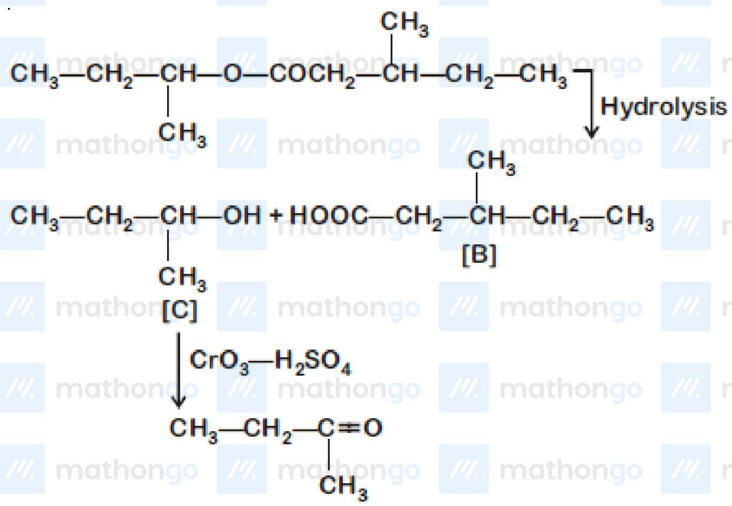
Answer Key

Q1 (C)	**Q2** (A, D)	**Q3** (D)	**Q4** (A)
Q5 (D)	**Q6** (C)	**Q7** (A)	**Q8** (D)
Q9 (69)	**Q10** (A)	**Q11** (C)	**Q12** (B)

Q1

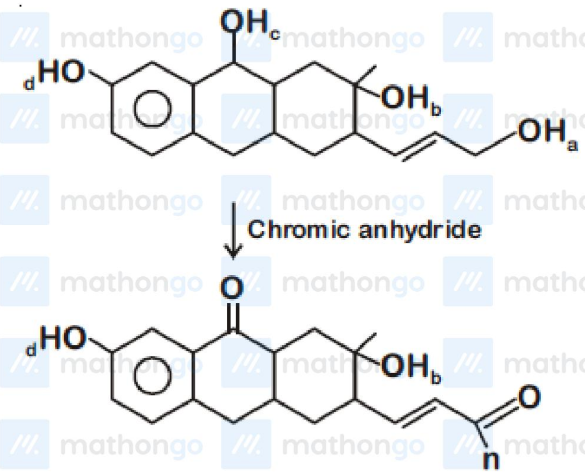


Q2



Also molecular formula of option (4) does not matches with (A)

Q3

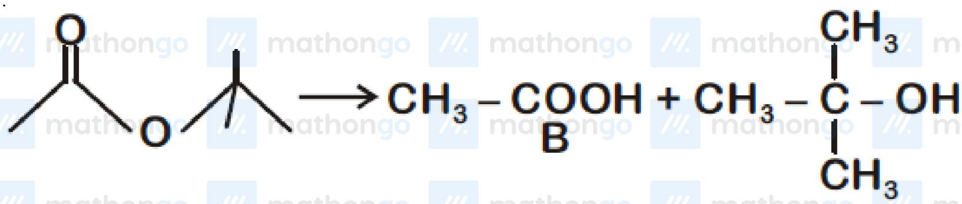


$\mathrm{H_b}$ and $\mathrm{H_d}$ can give CAN test

Q4

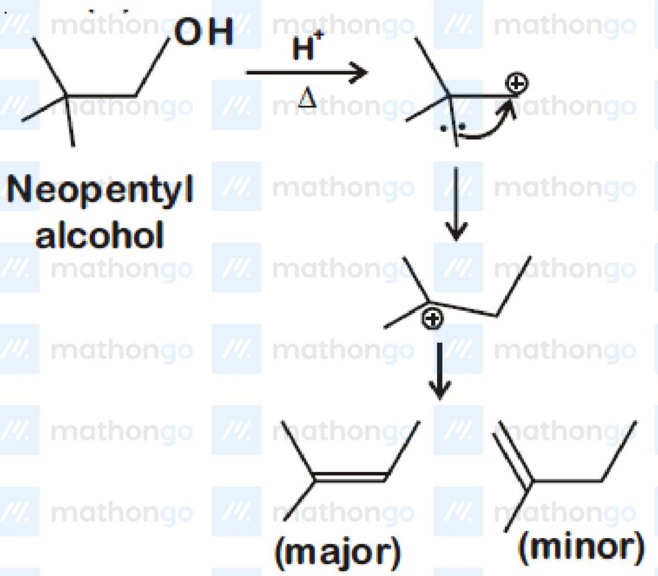
Ester $\xrightarrow{\text{hydrolysis}}$ $\underset{\text{(B)}}{\text{Acid}} + \underset{\text{(C)}}{\text{al cohol}}$

since C gives turbidity immediately with Lucas reagent (anhydrous $\mathrm{ZnCl_2}+$ conc. HCl). This means C should be 3° alcohol.

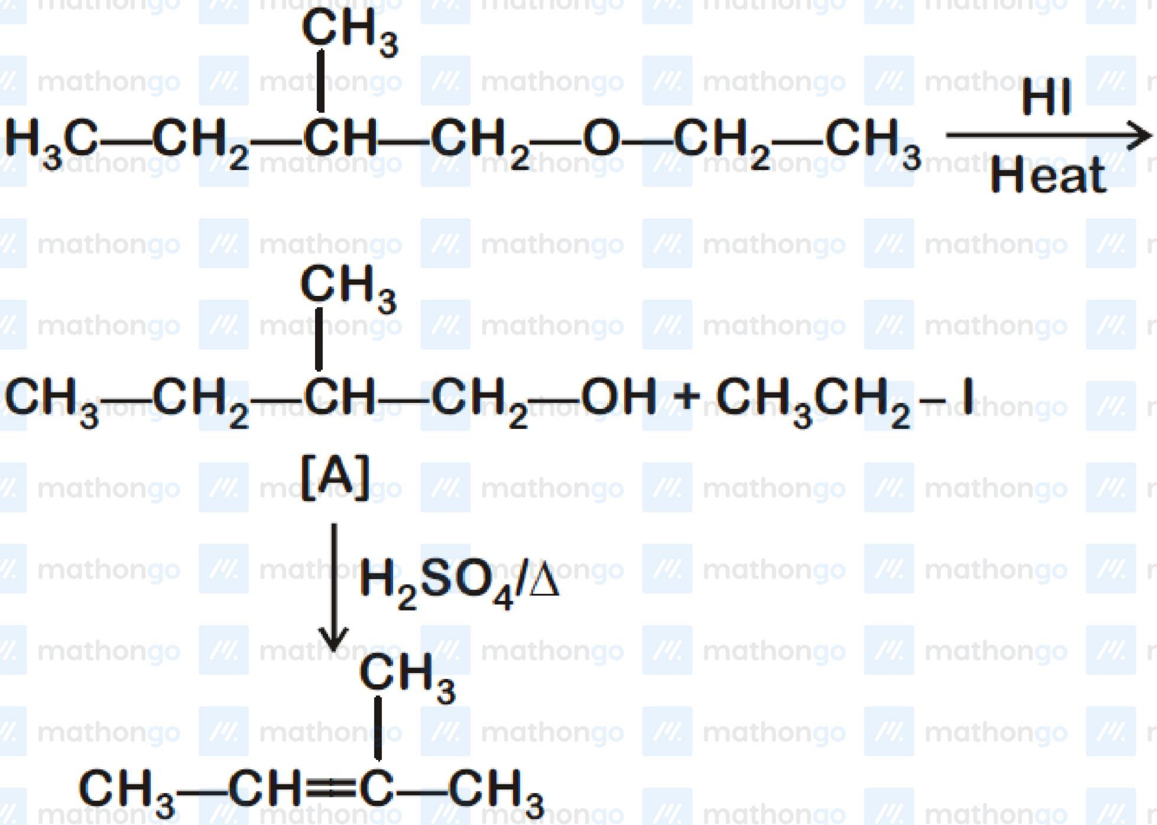


3° alcohol (C)

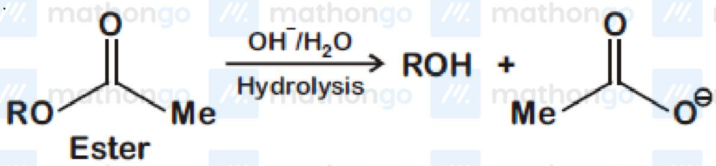
Q5



Q6

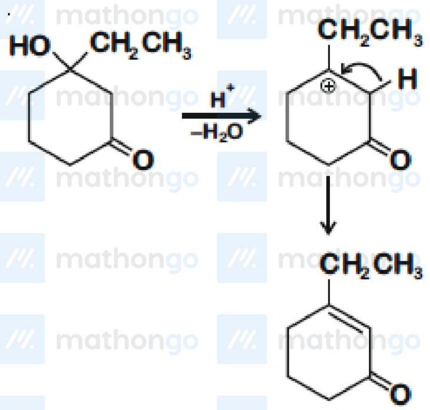


Q7

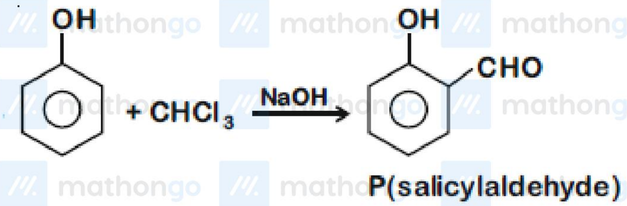


So, Ester is unstable in an aqueous basic solution and undergoes hydrolysis to give alcohol and carboxylate.

Q8

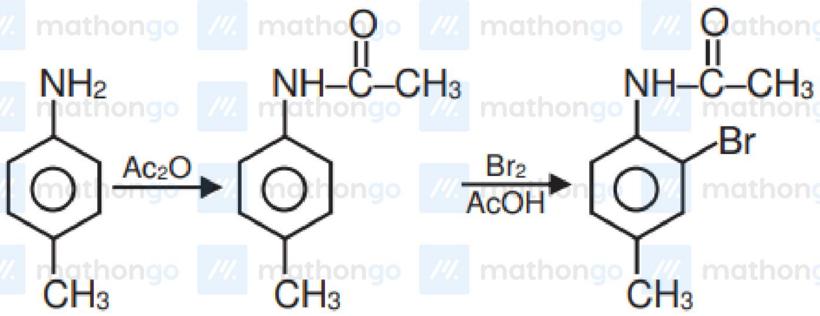


Q9

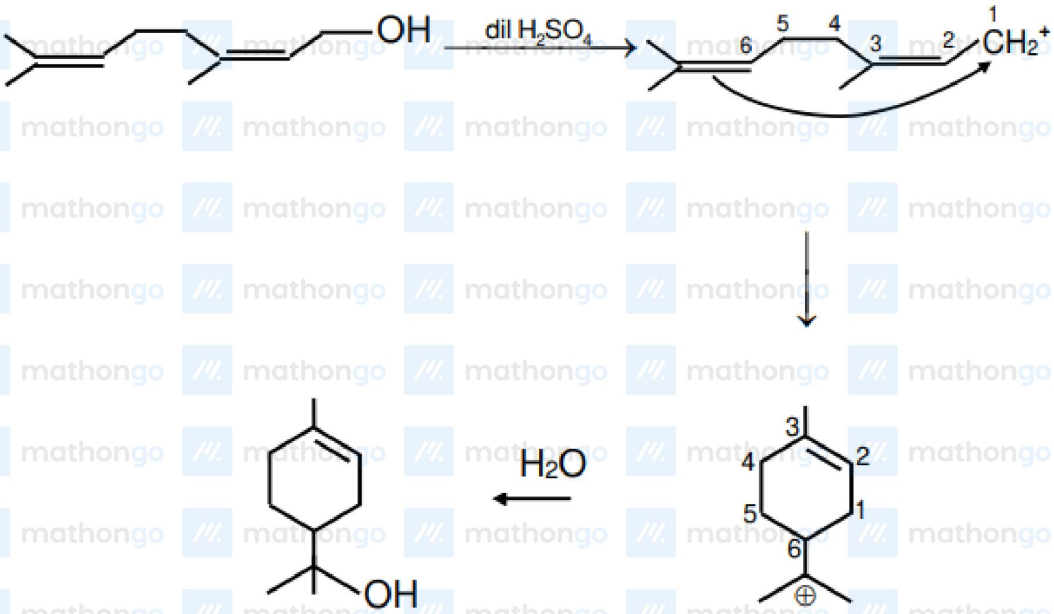


$\therefore$ mass % of C in P $= \frac{12\times7}{84+6+32} \times 100$
$= 68.85\%$
$\approx 69\%$

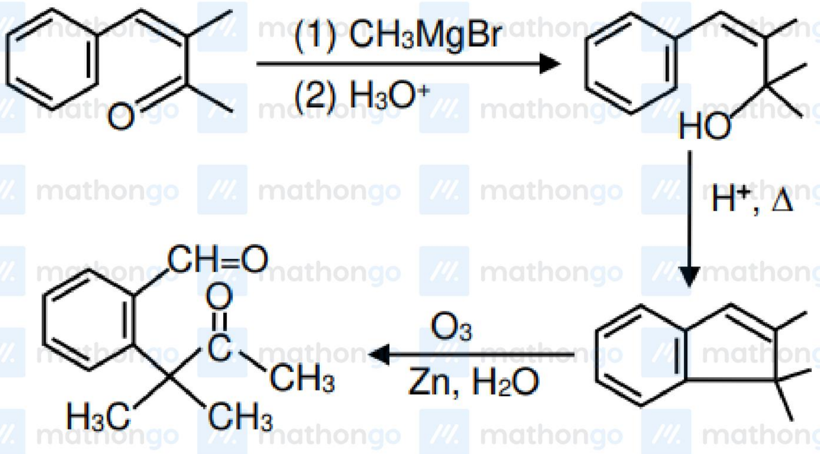
Q10



Q11


OH
dil H_2SO_4
1
2
3
4
5
6
CH_2^+
H_2O
OH
⊕

Q12


(1) CH_3MgBr
(2) H_3O^+
O
HO
H^+, Δ
O_3
Zn, H_2O
CH=O
O
C
CH_3
H_3C
CH_3